

02

LECTURE

BIOSTATISTICS AND EPIDEMIOLOGY



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DESCRIPTIVE STATISTICS

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- A collection of more than one figure is called statistics

- It's about collecting and processing information to stimulate additional questions, to provide a basis, and to assist in making more effective decisions

A. WHY STUDY STATISTICS?

- Data is collected everywhere, and we require statistical knowledge to make the information useful by processing and analyzing it.
- Statistical techniques are used to make professional and personal decisions.
- No matter your career, you need knowledge of statistics to understand the world and become conversant in your career.

B. APPLICATIONS OF STATISTICS

- In all fields of research, like marketing, accounting, business, quality control, politics, sports, health, education, etc.
- More on quantitative research

I. WHAT IS MEANT BY STATISTICS?

STATISTIC	STATISTICS
• A characteristic of a sample • A number used to communicate a piece of information • A number that is a part of a bigger picture • A datum that can be represented numerically	• The science of collecting, organizing, presenting, analyzing, and interpreting numerical data • More than the presentation of numerical information

II. TYPES OF STATISTICS

A. DESCRIPTIVE STATISTICS

- Can be used to organize data in meaningful forms.
- More on organizing and transforming organized data into something useful.
- Method of organizing, summarizing, and providing a description of data in an informative way.
 - Example: "A study conducted on a sample of patients revealed that 72% of participants were familiar with the concept of wearable health monitoring devices."
 - The statistic "72%" is only DESCRIBING a portion of a sample that is knowledgeable about wearable health devices.

B. INFERENCE STATISTICS

- Used to make decisions with a limited set of data.
- By taking a representative sample, we can make an inference that will hold true for the whole population.
- You just need to get a sample, create hypotheses and inferences, test, determine relationships, and make predictions.
- By taking a representative sample of blood specimens, the laboratory can obtain reliable estimates of the instruments' precision, allowing these findings to be extrapolated to a broader population of instruments.
- Enhances efficiency and affordability.

III. TYPES OF DATA

A. UNGROUPED DATA

- This is raw and unorganized information.

B. GROUPED DATA

- Organized or processed data, presented in a frequency table.

IV. POPULATION VS. SAMPLE

A. POPULATION

- Everyone you want to describe
- Expensive and time-consuming to measure everyone
- The complete set of all items, people, animals, or events that you want to study.
- It includes every single member that fits your research rules, not just a few.

B. SAMPLE

- Statistics = Sample
- The subset you actually measured.
- Representative of the population.
- It must be appropriately ratioed and generalizable to the entire population.
- A sample is often used in word problems.
 - Look for keywords such as "taken from" or "pulls the records of," "randomly selected from" etc.

	Population	Sample
Size	N	n
Mean	μ (mu)	$\bar{x}$
Variance	σ^2	s^2

**Standard
Dev**

σ

s

V. CENTRAL TENDENCY

- **Mean, median, mode**
- Where your data aggregates, or where most of your data is found.
- It is important to know which central tendency to use in a given situation or problem.

A. MEAN

- Arithmetic average.
- Only used for **intervals** and **ratios**.
- Very symmetrical, can be easily affected by extreme outliers.
- Every value is used.
- **NOT USED** for nominal/ordinal.
- **Formula:** Sum of all numbers $\div$ total number of data
- **NOTE:** means are pulled towards extreme outliers.
 - **Example:** The mean of a set of data with one very large number will be pulled towards that big number regardless of the other values; it will no longer speak for the rest of the values.
- If data is **ordinal** (I, II, 1, 2, 3), it cannot be averaged since it would not make sense and one level is higher than the other.
- If data is **nominal** (categorical), then you also cannot average that since it is separated into different categories.
- Properties:
 - Data must be in interval or ratio levels
 - All values are included in computing the mean
 - The mean is unique

- The sum of the deviations of each value from the mean is zero.

B. MEDIAN

- The middle value.
- The value that splits an ordered dataset exactly in half (**50% upper, 50% lower**).
- Used if the dataset has a lot of outliers and if data is **ordinal**.
- Data must be ordered first before finding the median (arranged from **least to greatest**).
- Easily found in **odd lists**: done by using the formula $(n+1) \div 2$ to find the middle number
- In **even lists**, find the two middles using the formula, then find the average of those two middles.
- **USED IF**: data is skewed due to extreme outliers, data is ordinal, or even in normal cases without any outlying extremes.
- **NOTE**: The median does not have to be a part of the data, especially in even lists. Whatever is the middle value (done by taking the average of the two middles) is the median.
- **ADDTL NOTE**: if the set is odd numbered, it means that the median is located among the dataset.

C. MODE

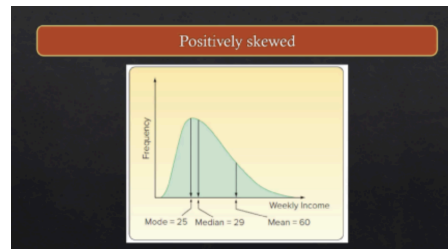
- Value that occurs most often.
- Used for **nominal** and **categorical** data.
- Can be **unimodal** (only one mode), **bimodal** (two modes), **multimodal** (more than two modes)
- Always found in your data set.
- In cases where there is no mode or more than one mode, better use the mean or median.
- No calculations needed, it mainly involves counting and comparing
- This is the only measure of central tendency valid for nominal (categorical data)

VI. DISTRIBUTION CURVES

A. NORMAL CURVE

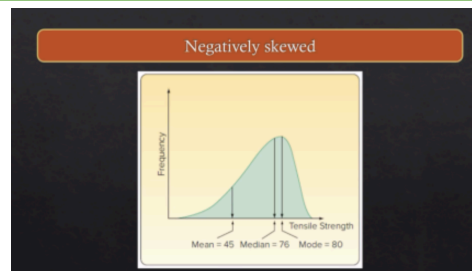
- Data is evenly halved.
- Symmetrical/ Bell curve
- mean = median = mode

B. POSITIVE SKEW



- The curve tilts to the left, forming a tail on the right side.
- Mean is greater than your median (mean > median > mode).
- **Formula**: mean - median = +value, meaning that there is a positive skew.
- **Example**: A positive skew is more favorable when a laboratory investigates a contamination case. The smaller the values, the lesser the likelihood of contamination is.

C. NEGATIVE SKEW



- The curve tilts to the right, forming a tail on the left side.

- Median is bigger than your mean (mode > median > mean).
- **Formula**: mean - median = -value, meaning there is a negative skew
- **Example**: A negative skew is more favorable when test results of an exam are released. Negative skew = more students passed or had good grades.

EXTRA NOTE!

It is not always ideal for a curve to be in normal distribution. Some scenarios favor a positive or negative skew.

VII. SPREAD

- How scattered your data is.
- Less scattered is more precise and accurate because it has a narrower range of values.
- More scattered, less reliable.

A. RANGE

- The difference between the biggest number and the smallest number of your dataset (maximum-minimum).
- Used if you want to have a rough sense of the spread.
- First look before computing variance.
- Lower range = more accurate since it has lesser variations of data
- Comparing the full spread of 2 small sets.
- Affected by extreme outliers.
- Only uses the highest and lowest value of the dataset

B. VARIANCE AND STANDARD DEVIATION

- Only done for interval/ratio.
- Variance is the average squared distance of each value from the mean.
- The SD is the square root.

- **NOTE:** the unit of a variance is always “unit squared” and is not yet the final answer that represents each mean. To get the final answer, you must first take the square root of it to get the actual unit of the value and to make it easier to compare.
- Much more precise than the range.

Sample Variance:

$$s^2 = \frac{\sum(x - \bar{x})^2}{n - 1}$$

Sample Standard Deviation:

$$s = \sqrt{\frac{\sum(x - \bar{x})^2}{n - 1}}$$

Population Variance:

$$\sigma^2 = \frac{\sum(x_i - \bar{x})^2}{N}$$

Population Standard Deviation:

$$\sigma = \sqrt{\frac{\sum(x_i - \bar{x})^2}{N}}$$

Legend:

x_i / x = value of one observation

N = population size

n = sample size

$\bar{X}$ = mean value of observations

s = sample standard deviation

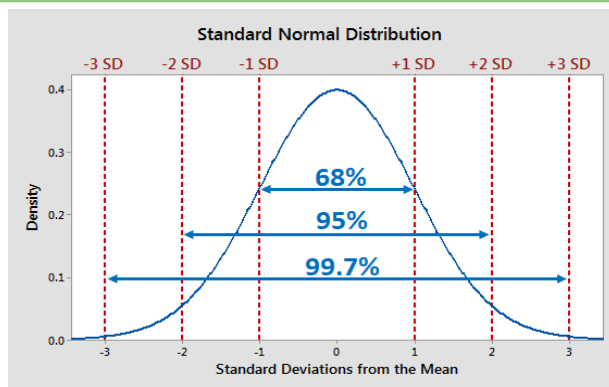
s^2 = sample variance

σ (**sigma**) = population standard deviation

σ^2 (**sigma squared**) = population variance

- How to easily solve for variance and standard deviation:
 1. Find the mean of all values by adding all sample values and divide by how many values are present
 2. Subtract the mean value of observations from each value ($x - \bar{x}$)
 3. Find the square of each ($x - \bar{x}$) and add them all together
 4. Divide the summation of all ($x - \bar{x}$)² with ($n - 1$) to find the variance
 5. Find the square root of the variance to get the standard variation.

C. EMPIRICAL RULE



- Most of the data (68%) is found between -1SD and +1SD.
- 95% between -2 and +2 sd.
- 99.7% between -3 and +3 sd .
- The remaining .3 are found outside the bounds as extreme outliers.
- Usually used in the laboratory to check for quality control of instruments. (Westgard-style control rule)
 - **Example:** You are doing quality control for four straight days, and each result has consistently been between -1 and +1. On the fifth day, it reached -3, which indicates that something is wrong with the machine or with the reagents.
 - Machines usually give out the data, and it is up to us to interpret the data. Mostly used to determine the normal values in laboratory tests.

VIII. POSITION

- **Percentiles, quartiles, deciles**
- Your placement as a whole.
- Where your data is located in comparison to the entire set.
- **When to use:** ranking an individual result against a reference group (growth charts,

CD4 counts), describing skewed data without assuming normality, setting service-level benchmarks (e.g., “90% of STAT results within 30 minutes”), and flagging outliers via the interquartile range (IQR = Q3 - Q1).

- **Formula:** Position of Pk = $(n+1)P/100$
- If quartile: k= 25, 50, 75
- If decile: k= 10, 20, ..., 90
- If percentile: k= 1-99

A. QUANTILES

- Divide a set of observations into 4 equal parts.
- Divides a dataset into 4 parts, with 3 main divisions (Q1, Q2, Q3).
- **Q1**= 25th percentile
- **Q2**= median (50th percentile)
- **Q3**= 75th percentile
- **IQR (Interquartile Range)**
 - Measures the spread of the middle 50% of a data set.
 - The range from Q1 to Q3.
 - **Formula:** Q3- Q1
- **Upper Fence**
 - A calculated boundary value used to identify high outliers in a dataset.
 - **Formula:** Q3 + 1.5 (IQR)
- **Lower Fence**
 - A calculated limit in statistics used to find abnormally low data points, known as outliers.
 - **Formula:** Q1 - 1.5 (IQR)

B. PERCENTILE

- A percentile score or centile, is a score below which a given percentage k of scores in its frequency distribution falls or a score at or below which a given percentage falls.
- Divides dataset into 100 parts (P1 to P99).
- Most granular divisions, more accurate

C. DECILE

- Divided into 10 parts (D1–D9).
- D9 is the 90th percentile.

Example: If, in an exam, you got a 90 out of 100, but 18 out of 20 of your classmates got a score above 90, then it means that you are on the lower end of the percentile, since you have one of the lowest scores.

IX. TABLES AND GRAPHS FOR ORGANIZING RAW DATA

A. FREQUENCY DISTRIBUTION TABLE

- Done to show the general pattern of the data, identify where values tend to concentrate, and show extreme and outlying values.
- Grouping of data into **mutually exclusive** and **collectively exhaustive** classes, showing the number of observations in each class.
- Mutually exclusive:
 - Data only belongs to one category (if it is pediatric data, it only belongs to the pediatric category).
- Collectively exhaustive:
 - Each data point has a group to which it belongs
 - The total number of all data ensures that every test result is represented in the table.

Patient Age Group	Number of Results
Neonates	40
Pediatric	52
Adult	45

Elderly	43
TOTAL	180

B. RELATIVE CLASS FREQUENCIES

- Captures the relationship between a class frequency and the total number of observations.
- Found by dividing each number of results by the total.

Patient Age Group	# of Results	Relative Frequency	Found by
Neonates	40	0.222	40/180
Pediatric	52	0.289	52/180
Adult	45	0.250	45/180
Elderly	43	0.239	43/180
TOTAL	180	1.000	

C. CONSTRUCTING FREQUENCY DISTRIBUTIONS

Glucose Levels Tested (in mg/dL)			
105	140	98	132
120	100	138	118
92	85	115	105
135	125	130	152
110	105	112	115
88	118	112	125
150	160	105	110
95	112	140	98
130	108	90	135
115	142	128	108
Minimum: _____			
Maximum: _____			

- Decide on the number of classes
 - Classes = groups
 - Follow the rule “2 raised to k”
 - In our example, we have to think of a number that can be substituted

to k in order to get a number that is equal to or greater than the number of values

- In our example:
 - If k=5, it is wrong since $2^5 = 32$ which is less than 40 and it means that not all values will have a corresponding class
 - If k=6, it is correct since $2^6 = 64$ since it is greater than 40

- Determine the class interval

- Class interval = limits of each group
- Must be the same in all classes, Must cover the distance between the minimum value and the maximum value
- Follow this formula:

$$i \geq \frac{\text{Maximum Value} - \text{Minimum Value}}{k}$$

- In the example, we have 85 as the minimum and 160 as the maximum. Thus:
 - $(160-85) \div 6 = 12.5$
- Usually, the intervals are mostly rounded to more convenient numbers such as multiples of 5, 10 and 15. Thus:
 - $12.5 \rightarrow 15$

- Set the individual class limits

Number of classes = 6
Class interval = 15

Minimum: 85
Maximum: 160

Classes
80 up to 95
95 up to 110
110 up to 125
125 up to 140
140 up to 155
155 up to 170

- **Note:** Each observation must belong to only one category, meaning overlapping or unclear class limits must be avoided. The minimum limit should be based on the minimum value. Then, check if the maximum value will fit into a class.
- Example: If the minimum value is 85, start at 80. Adding the class interval of 15 equals 95. Therefore, all numbers from 80 up to 95 belong to the first class/group.
- Since 95 is the last number of the first class/group, the next class starts at 95 as the lower limit.
- The same process continues for the subsequent classes until the ideal number of classes, which is 6, is achieved.
- Here, 'up to' refers to numbers spanning from 80 up to 95.
 - Specifically, it denotes the range $80 \leq x < 95$, which means x is greater than or equal to 80, but strictly less than 95.
- If the number reaches 95, it would then be considered to another group/class
- Cannot write "80-95" and "95-110" since it is ambiguous since it has both 95 in its limits.
- Cannot change the class limit into "80-94" and "95-110" since there is a possibility that a value of 94.5 would appear
- It wouldn't be covered with the class limit's ranges since the 1st group is only until 94, and the 2nd group starts with 95, leaving 94.5 to neither of the class limits.

4. Tally

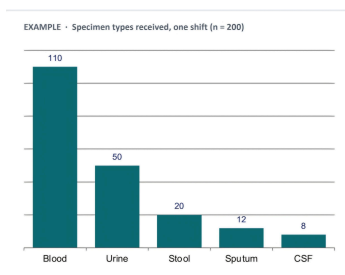
105	140	98	132
120	100	138	118
92	85	115	105
135	125	130	152
110	105	112	115
88	118	112	125
150	160	105	110
95	112	140	98
130	108	90	135
115	142	128	108

Classes	Frequency
80 up to 95	
95 up to 110	-
110 up to 125	- -
125 up to 140	-
140 up to 155	
155 up to 170	

- Tally the observations into classes and determine the no. of observations in each class.
- The number of observations in each class is called class frequency.
- Now you can observe that the glucose level results are mostly confined between 95 to 140 mg/dL.
- You can also find the class limit per class, which is done by adding the upper and lower limit and divide it by 2.

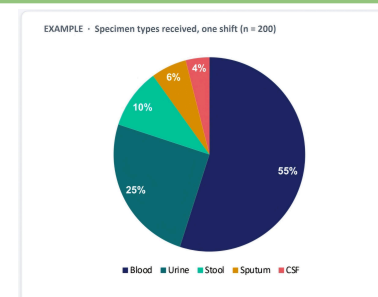
X. TABLES AND GRAPHS FOR COMPARING CATEGORIES

A. BAR GRAPH



- (Nominal and Ordinal)
- Bars show the frequency or count of each category. Bars are drawn separated by gaps — the gap is a visual reminder that the categories are distinct, not continuous.
- **When to Use:**
 - Comparing frequencies across nominal or ordinal categories
 - Ranking or comparing several groups side by side
 - Precise comparison matters more than "share of the whole"
- Not for continuous numeric data .

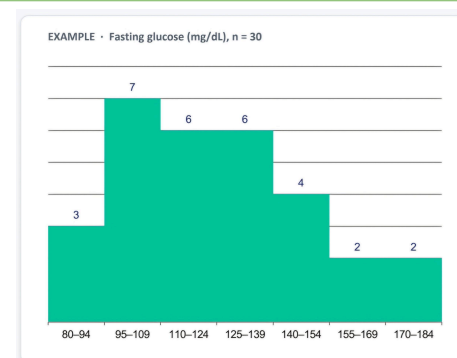
B. PIE CHART



- (Nominal and Ordinal)
- A circle divided into slices sized by each category's share of the total. Built to emphasize part-of-a-whole — the slices should sum to 100%.
- **When to Use:**
 - Showing proportion or percentage composition of a whole.
 - Few categories — ideally 5 or fewer, so slices stay readable.
 - The "share of total" story matters more than exact ranking.
- Use specifically if you want to illustrate a class' part of a whole.

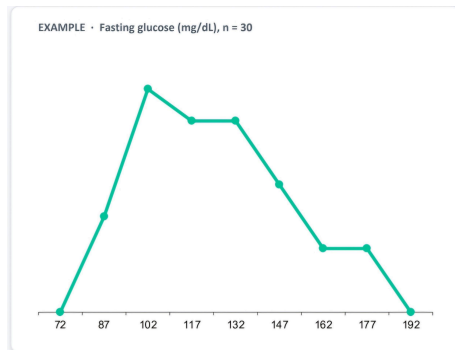
XI. TABLES AND GRAPHS FOR SHOWING DISTRIBUTION SHAPE

A. HISTOGRAM



- (Interval/Ratio (Continuous))
- Bars show the frequency of continuous numeric data grouped into class intervals.
- Bars touch, with no gaps, because the underlying variable is continuous.
- **When to Use:**
 - Visualizing the shape of a continuous distribution (symmetric, skewed, bimodal).
 - A reasonably large sample — generally $n \geq 20-30$ — to form meaningful classes.
 - A quick visual check before computing skewness or choosing mean vs. median.
- Its lack of gaps represent that the data it shows is continuous, which differs with the bar graph.

B. FREQUENCY POLYGON



- (Interval/Ratio (Continuous))
- A line graph formed by connecting the midpoint of each class interval at its frequency, then dropping to zero at both ends — an alternative way to show the same shape a histogram shows.
- **When to Use:**
 - Comparing two or more distributions on the same axes.
 - Emphasizing overall trend/shape over individual bar heights.

- Continuous numeric data, using the same class intervals as a histogram

- Two groups' polygons can share one chart — much easier to compare than two overlapping histograms.

C. STEM AND LEAF PLOT

EXAMPLE · Fasting glucose (mg/dL), $n = 30$ · Stem = tens digit, Leaf = ones digit

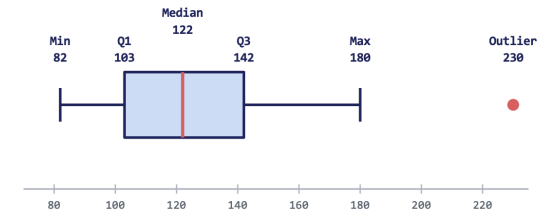
Stem	Leaves
8	2 8
9	1 5 6 8
10	1 3 4 7
11	0 2 3 6 9
12	2 5 7 8
13	1 4 7
14	0 2 5
15	0 8
16	5
17	2
18	0

- (Interval/Ratio)
- Sorts data by leading digit(s) — the “stem” — and lists the trailing digit — the “leaf” — for each value. Shows the shape of the distribution while preserving every original value.
- **When to Use:**
 - Small to moderate datasets (roughly $n < 50$).
 - You need the shape AND the exact values, not just bin counts.
 - Quick-by-hand data exploration without software.
- Unlike a histogram, every original data point can still be read directly off a stem-and-leaf plot — nothing is lost to binning.

XII. SHOWING SPREAD & OUTLIERS

A. BOX-AND-WHISKER PLOT

EXAMPLE · Fasting glucose (mg/dL), $n = 31$ (incl. 1 severe result)

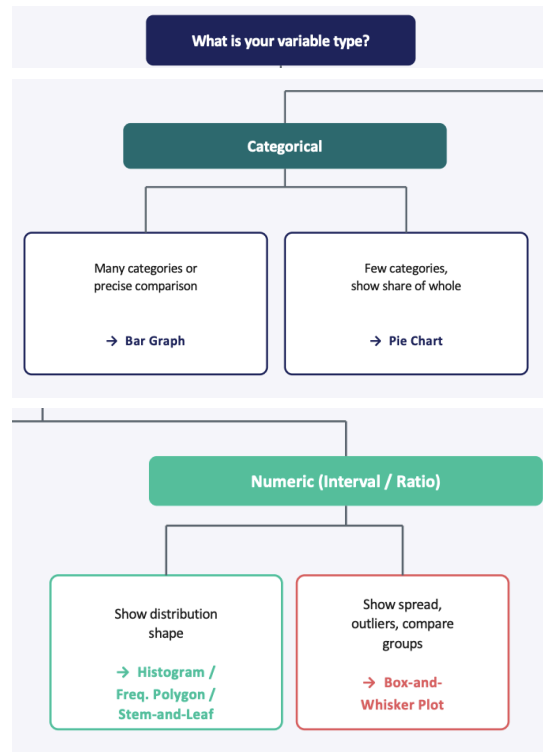


- (Interval/Ratio)
- A five-number summary — minimum, Q1, median, Q3, maximum — drawn as a box with whiskers. Built to show spread, skew, and outliers at a glance, and to compare groups side by side.
- **When to Use:**
 - Comparing spread and center across two or more groups.
 - Identifying outliers using the $1.5 \times \text{IQR}$ rule.
 - Data are skewed and you want a summary that resists outliers.
- The box covers the middle 50% of the data (the IQR). Points beyond $1.5 \times \text{IQR}$ from the box are plotted individually as outliers — exactly what we flagged below.

XIII. WHICH DISPLAY FITS WHICH VARIABLE?

Display	Categorical (Nominal/Ordinal)	Numeric (Interval / Ratio)	Shows Exact Values?	Primary Purpose
FDT	✓	✓	No (grouped)	Organize raw data into classes
Bar Graph	✓	✗	No	Compare category frequencies
Pie Chart	✓ (few cats.)	✗	No	Show proportion of a whole
Histogram	✗	✓	No (binned)	Show shape of a continuous distribution
Frequency Polygon	✗	✓	No (binned)	Compare shapes of 2+ distributions
Stem-and-Leaf Plot	✗	✓ (n < ~50)	Yes	Show shape AND exact values
Box-and-Whisker Plot	✗	✓	No (5-number summary)	Show spread, skew, outliers; compare groups

XIV. WHICH DISPLAY SHOULD I USE?



XV. SUMMARY

A. MEAN VS MEDIAN VS MODE

	Mean	Median	Mode
Best data type	Interval/ratio	Ordinal, interval, ratio	Nominal
Uses every value?	Yes	No, only middle	No, only frequency
Affected by outliers?	Yes—strongly	No—resistant	No
Always exists/unique?	Yes, unique	Yes, unique	May be none, 1, or several
Best when?	Symmetric distribution	Skewed distribution / outliers	Describing categories or a peak

B. RANGE VS VARIANCE VS SD VS EMPIRICAL RULE

	Range	Variance	S.D.	Empirical rule
What it tells you	Total spread (2 pts. only)	Average squared distance from mean	distance from mean, original units	% of data within $\pm 1/2/3$ SD
Units	same as	Squared units	Same	Percent

	data		as data	
Sensitive to outliers?	Very	Yes	Yes	Depends on normality
Requires normal shape/curve?	No	No	No	Yes (approximately)

C. PERCENTILE VS QUARTILE VS DECILE

	Quartile	Decile	Percentile
Divides data into	4 equal parts	10 equal parts	100 equal parts
Notation	Q1, Q2, Q3	D1–D9	P1...P100
Equivalent to	Q2=P50=median	D5=P50=median	P25=Q1, P50=Q2, P75=Q3
Typical lab use	IQR/ outlier detection	Workload or turnaround time benchmarking (what the TAT should be)	Ranking one patient result vs a reference group

WORKSHEET ANSWERS

Scenario 1.1 – Donor Hemoglobin Screening

- A city blood bank is revising its donor-eligibility cutoff and wants a snapshot of typical hemoglobin (Hgb) levels among healthy donors. A medical laboratory scientist randomly pulls the records of 15 donors from last month's donation log.
- Data:** Hemoglobin (g/dL), n = 15

Donor #	Hgb (g/dL)
1	13.2
2	14.5
3	13.8
4	15.1
5	14
6	13.5
7	14.8
8	13.9
9	14.6
10	14.2
11	13.6
12	14.4
13	13.7
14	14.1
15	13.3

- Mean:** 14.05 g/dL
- Sum of all values ÷ Total number of values
 - $(13.2 + 14.5 + 13.8 + 15.1 + 14 + 13.5 + 14.8 + 13.9 + 14.6 + 14.2 + 13.6 + 14.4 + 13.7 + 14.1 + 13.3) \div 15 \approx 14.05 \text{ g/dL}$
- Median:** 14.0 g/dL
- The middle value when all values are sorted from smallest to largest.
 - 13.2, 13.3, 13.5, 13.6, 13.7, 13.8, 13.9, **14.0**, 14.1, 14.2, 14.4, 14.5, 14.6, 14.8, 15.1
 - With 15 values, the 8th value sits exactly in the middle — 14.0 g/dL
- Mode:** none

- All values appear only once — there are no repeated numbers.

- Sample/Population?:** Sample
 - The group of 15 people was randomly selected from a larger group — all donors from last month. Since this is only a portion of the total group, it is a sample, not the full population.

Scenario 1.2 – Blood Type Inventory Planning

- The blood bank's inventory manager wants to know which ABO/Rh blood type is most in demand this month, based on 50 walk-in donors, to plan storage and outreach.
- Data: Blood type frequency, n = 50 donors

Blood Type	Frequency
O+	20
A+	12
B+	10
AB+	3
O-	3
A-	1
B-	1
AB-	0

- Modal Blood Type:** O+
- Frequency of Mode:** 20
 - O+ is the most frequent category (20 occurrences), so it is the modal blood type.
- Can mean/median be computed?** No
 - Blood type is nominal data** — categorical labels with no numerical value or order. Mean and median only apply to quantitative (interval/ratio) data that can be summed, measured, or ranked sequentially.

Scenario 2.1 – Urine Culture Colony Counts

- In the microbiology section, a medical technologist records colony-forming unit (CFU/mL) counts from 20 urine cultures submitted with suspicion of urinary tract infection (UTI).
- Data: Colony count (CFU/mL), n = 20

Sample #	CFU/mL
1	10000
2	15000
3	8000
4	12000
5	20000
6	9500
7	11000
8	100000
9	85000
10	13500
11	10500
12	9000
13	14000
14	95000
15	12500
16	11500
17	10200
18	13000
19	9800
20	120000

- Mean:** 29,475 CFU/mL (sum of all values ÷ total count)
- Median:** 12,250 CFU/mL
 - with two middle values: 12,000 and 12,500; average = $(12,000 + 12,500) \div 2 = 12,250$
- Skew Direction:** Positive (+)
 - Skew is positive because: Mean > Median. This occurs because a few extremely large values pull the mean upward, creating a longer tail extending toward the right side of the distribution.

Scenario 2.2 – Mock Licensure Board Exam Scores

- An MLS review center gives a mock board exam (%) to 25 graduating students two weeks before the actual licensure exam, after an intensive review program.
- Data: Mock exam score (%), n = 25

Student #	Score (%)
1	88
2	90
3	92
4	85
5	91
6	89
7	93
8	87
9	90
10	86
11	88
12	91
13	84
14	92
15	89
16	90
17	45
18	88
19	91
20	87
21	90
22	93
23	86
24	89
25	60

- Mean:** 86.16% (sum of all values ÷ total count)
- Median:** 89.0% (middle value)
- Mode:** 90.0% (appeared 4 times — more frequently than any other value)
- Skew Direction:** Negative (–)
 - Skew is negative because: Mean < Median < Mode. This pattern occurs due to a few extremely low values, which pull the mean downward and create a longer tail extending toward the left side of the distribution.

Scenario 3.1 – Comparing Glycemic Control Between Two Drug Regimens

- A hospital laboratory is asked to help evaluate two treatment regimens for Type 2 diabetes. Fasting blood glucose (mg/dL) is measured in 10 patients on Regimen A and 10 different patients on Regimen B.
- Data: Fasting blood glucose (mg/dL)

Patient #	Regimen A (mg/dL)	Regimen B (mg/dL)
1	117	86
2	122	136
3	115	101
4	127	156
5	119	91
6	125	146
7	116	96
8	123	141
9	118	94
10	121	151

	Regimen A	Regimen B
Mean	120.3 mg/dL	119.8 mg/dL
Range	12 mg/dL	70 mg/dL
Sample Variance	15.79 mg/dL	804.4 mg/dL
Sample SD	3.97 mg/dL	28.3 mg/dL

- Range** = Highest value - Lowest value
 - Example: Regimen A = $127 - 115 = 12$
- Sample Variance**
 - Subtract the mean from each value, then square the result.
 - Example: $117 - 120.3 = -3.3$
 $\rightarrow (-3.3)^2 = 10.89$
 - Sum all squared differences.
 - $10.89 + 2.89 + 28.09 + 44.89 + 1.69 + 22.09 + 18.49 + 7.29 + 5.29 + 0.49 = 142.10$

- Divide the total by $n - 1$ (sample size minus one).
 - $142.10 \div (10 - 1) = 142.10 \div 9 = 15.79$

- **Standard Deviation**

- Square root of the sample variance
- $\sqrt{15.79} = 3.97$

Scenario 3.2 - Establishing a Serum Calcium Reference Range

- The clinical chemistry section is validating a reference interval for serum calcium using 200 healthy adult volunteers. The resulting distribution is approximately normal, with a mean of 9.5 mg/dL and a standard deviation of 0.4 mg/dL.

Given

Mean (μ), mg/dL	9.5
Standard Deviation (σ), mg/dL	0.4
Patient's result, mg/dL	10.7

68% range (+1 SD): low	9.1
68% range (+1 SD): high	9.9
95% range (+2 SD): low	8.7
95% range (+2 SD): high	10.3
99.7% range (+3 SD): low	8.3
99.7% range (+3 SD): high	10.7
Within +2 SD reference interval? Y/N	N (8.7 > 10.3)

- The data follows a normal distribution, so the **Empirical Rule** applies.
- Calculate each interval by adding/subtracting the SD multiple from the mean:
 - **± 1 SD (68%):** Subtract SD for lower bound, add SD for upper bound.
 - **± 2 SD (95%):** $SD \times 2 = 0.4 \times 2 = 0.8$. Subtract/add 0.8 to the mean.
 - **± 3 SD (99.7%):** $SD \times 3 = 0.4 \times 3 = 1.2$. Subtract/add 1.2 to the mean.
- Since the observed value (10.7) falls outside the ± 2 SD range (8.7–10.3), it is not within the 95% reference interval.

Scenario 4.1 - Ranking a CD4 Count

- At an HIV treatment clinic, CD4 counts (cells/mm³) are recorded for 40 patients during routine immune monitoring. One patient of interest has a CD4 count of 350 cells/mm³, which corresponds to the 25th percentile of the clinic's patient distribution.

Rank	CD4 (cells/mm ³)
1	150
2	170
3	185
4	200
5	215
6	230
7	245
8	260
9	280
10	350
11	360
12	375
13	390
14	410
15	425
16	440
17	460
18	480
19	500
20	520

21	540
22	560
23	580
24	600
25	620
26	640
27	660
28	680
29	700
30	720
31	740
32	760
33	780

34	800
35	820
36	840
37	860
38	880
39	900
40	920

- **Patient's percentile rank (of 350):** 25th
- **Interpretation:** low
 - To get the patient's percentile rank based on the scenario given, we simply just follow the formula: (number of values below 350 / 40) x 100.
 - 40 is our **n** in this situation, basically representing the number of ranks per patient (which is 40).
 - There is little need to solve this as the worksheet explicitly states that the patient belongs to the **25th percentile**.
 - It is interpreted as **low**, as the patient falls in the 25th percentile out of 100.
 - This means that the patient has a CD4 count lower than 75% of patients CD4 counts presented in the table.

Scenario 4.2 - Cholesterol Screening and Outlier Detection

- During a workplace wellness screening, total cholesterol (mg/dL) is measured for 24 employees.
- **Data: Total Cholesterol (mg/dL), $n = 24$, already sorted.**

Rank	Cholesterol (mg/dL)
1	150
2	155
3	160
4	162
5	165
6	168
7	170
8	172
9	175
10	178
11	180
12	182
13	185
14	188
15	190
16	192
17	195
18	198
19	200
20	205
21	210
22	215
23	220
24	280

- **Q1:** 168.5 mg/dL
 - **Position Formula: $(n+1)25/100$**
 - $(24+1)25/100 = 6.25$ th Position
 - Q1 lies 0.25 of the way between the 6th value (168) and the 7th value (170):
 - Difference: $170 - 168 = 2$
 - Portion: $2 \times 0.25 = 0.5$
 - $Q1 = 168 + 0.5 = 168.5$
- **Q2 (Median):** 183.5 mg/dL
 - We start by finding Q2 (the median). Since there are **24 values** (an even number), the **median is the average of the 12th and 13th values**:
 - $(182 + 185) / 2 = 183.5$
- **Q3:** 199.5
 - $(24 + 1) \times 75 \div 100 = \mathbf{18.75}$ th position
 - This means Q3 lies three-quarters of the way from the 18th value to the 19th value
 - **Difference:** $200 - 198 = 2$

- **Portion:** $2 \times 0.75 = 1.5$
- **Q3** = $198 + 1.5 = 199.5$
- **IQR (Needed if there are outliers):** 31 mg/dL
 - **Q3 - Q1** = $199.5 - 168.5 = 31$
- **Upper Fence (To determine Outlier):** 246 mg/dL
 - **Q3 + 1.5 x IQR** = $199.5 + 1.5 (31) = 246$
- **Outlier(s) present? Value(s):** Yes. 280 mg/dL

Explanation:

- To find the upper fence, multiply the IQR by 1.5 and add it to Q3:
 - $199.5 + (1.5 \times 31) = \mathbf{246}$.
- Any value higher than 246 is considered an outlier.
 - Since **280** is greater than 246, it is an outlier.

Scenario 4.3 – Benchmarking STAT Test Turnaround Time

- The laboratory's quality team is setting a service-level target for STAT troponin testing (used to rule out heart attack in the ER). They collect turnaround time (TAT, in minutes) for 50 STAT troponin orders and want at least 90% of results released within a target time.
- **Concept Focus:** the target is expressed as the 9th decile (D9 / P90) of the TAT distribution, not the mean.

Order #	TAT (minutes)
1	22
2	25
3	28
4	30
5	31
6	33
7	35
8	36
9	38
10	40
11	41
12	42
13	43
14	44
15	45
16	46
17	47
18	48
19	49
20	50
21	28
22	30
23	32
24	34
25	36
26	38
27	40
28	42
29	44
30	46
31	48
32	50
33	52
34	54
35	56
36	58

37	60
38	62
39	64
40	66
41	70
42	75
43	80
44	90
45	100
46	120
47	140
48	150
49	160
50	180

Mean TAT: 57.56 mins.

90th Percentile (P90/D9): 118 mins.

- **$(n+1)90/100$**
- $(50+1) 90/100 = 45.9$ th Position
- 45.9th Position (Between 100 and 120)
- **Difference:** $120 - 100 = 20$
- **Portion:** $20 \times .9 = 18$
- **P90** = $100 + 18 = 118$

Does mean alone reveal the slow tail? (Y/N): No.

Explanation:

- The mean TAT is taken using the formula you use for solving means in general, which is to add all values then divide it by n.
- In this situation, n = 50 therefore we divide the total by 50. Doing so will give us **57.56 minutes** as our answer.
- To find the 90th percentile or our P90/D9, we use the formula $L_p = (n+1)(P/100)$.

- Plugging in our values, that's $L90 = (50+1)(90/100)$. Next, it goes to $(0.90)(51) = 45.9$ (Position).
- This tells us that our P90 falls between the 45th and 46th values, which is 100 and 120 respectively.
- Now, using the percentile formula which is $P = X + P \text{ as decimal } [(Y - X)]$.
- Y and X are placeholder letters that signify the two values we got from calculating the P90 location.
 - $X = 100$
 - $Y = 120$
- Plugging in the values into our formula earlier, we get an equation of $P = 100 + 0.90(120-100)$, which gives us the answer of **118 minutes**.
 - Make sure to follow PEMDAS to arrive at the same answer.
- Does mean alone reveal the slow tail?
 - The answer is **no**.
 - As you can see, the mean only goes to 57.56 minutes, but according to our data set, the upper end stretches to 180 minutes.
 - Our P90, however, is closer to the upper end than the mean.
 - Therefore, using the P90 is more useful as a basis for finding the location of the slow tail.